



LUND
UNIVERSITY

Numerical Analysis for E3
Numerical Analysis, Centre for Mathematical Sciences

Achim Schroll, March 2003

Homework assignment 2

Delivery date: Tuesday, April 1st 2003, no later than 18:00h.

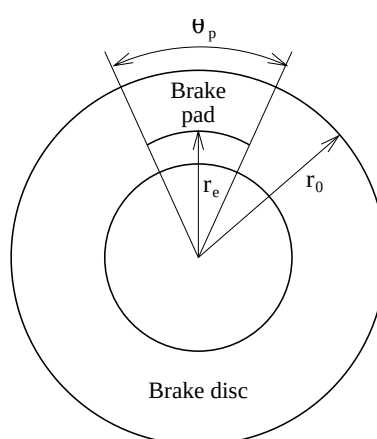
Topics of this assignment: Numerical integration: quadrature problem; Simpson's integration rule; Gauß integration.

1. Integration of tabulated data. To simulate thermal properties of a brake system, see the figure below, an average temperature of the breaking block is needed. This average temperature is given by

$$T_{\text{avg}} = \frac{\int_{r_e}^{r_0} T(r) r \, dr}{\int_{r_e}^{r_0} r \, dr}$$

where $T(r)$ is the temperature at distance r from the brake center. We know that $r_e = 9.38$ cm and $r_0 = 14.58$ cm. $T(r)$ is given for some values of r in the table below. The data were obtained by numerical solution of the heat equation.

r (cm)	$T(r)$ (°C)
9.38	338
9.90	423
10.42	474
10.94	506
11.46	557
11.98	573
12.50	601
13.02	622
13.54	651
14.06	661
14.58	671



Use these values to find an approximation of the average temperature T .

2. Simpson quadrature.

Task 2a): Derive Simpson's rule

$$\int_0^1 f(x) \, dx = \frac{1}{6} (f(0) + 4f(1/2) + f(1)) ,$$

by integrating the second degree interpolating polynomial.

Hint: Remember the Lagrange interpolating polynomial

$$P_2(x) = f(0)L_1^2(x) + f(1/2)L_2^2(x) + f(1)L_3^2(x) .$$

Task 2b): Apply Simpson's rule to compute the integral

$$\int_0^{\frac{1}{2}} x^2 \arccos x \, dx$$

with different step sizes:

$$h = 2^{-i}$$

with $i = 1, 2, 3, 4, 5, 6$. Compare your results with the exact solution which is 0.04934910257597 and plot a diagram of the step size versus error. Use a loglog diagram to easily check the order of the method graphically.

3. Gauss quadrature.

Task 3a): Derive the 4th. order Gauss formula.

Hint: The knots c_j are given by $2c_j - 1 = \gamma_j$, where γ_j are the roots of the Legendre polynomial $P_2(t) = \frac{1}{2}(3t^2 - 1)$. The weights b_j can be obtained either by integrating the Lagrange polynomials defined by the knots (as outlined in the lecture), or from the order conditions, Theorem 32. (See also Section 1.4.2.)

Task 3b): Test your formula for the integral $\int_0^{\pi/2} t \cos t + e^t \, dt$, the well known test example from the lecture.